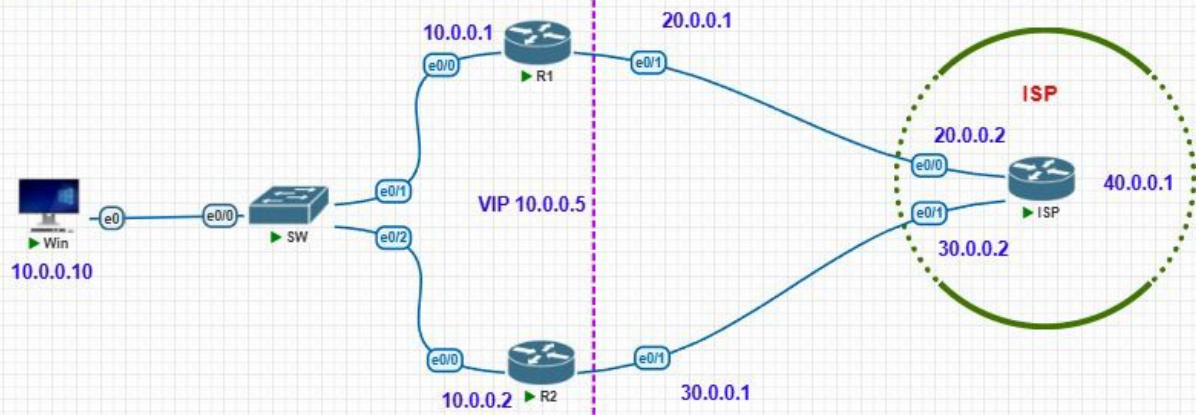




```
R1
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#int e0/0
R1(config-if)#standby 1 ip 10.0.0.7
R1(config-if)#no sh
R1(config-if)#ex
R1(config)#
*Aug 15 15:15:15.480: %HSRP-5-STATECHANGE: Ethernet0/0 Grp 1 state Standby -> Active
R1(config)#
R1(config)#
R1(config)#
R1(config)#do sh standby
Ethernet0/0 - Group 1
  State is Active
    2 state changes, last state change 00:01:36
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.416 secs
  Preemption disabled
  Active router is local
  Standby router is 10.0.0.2, priority 100 (expires in 9.216 sec)
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R1(config)#
```

```
R2
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#do sh standby
Ethernet0/0 - Group 1
  State is Standby
    1 state change, last state change 00:01:21
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.080 secs
  Preemption disabled
  Active router is 10.0.0.1, priority 100 (expires in 9.984 sec)
  Standby router is local
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
```

```
QEMU (Win)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

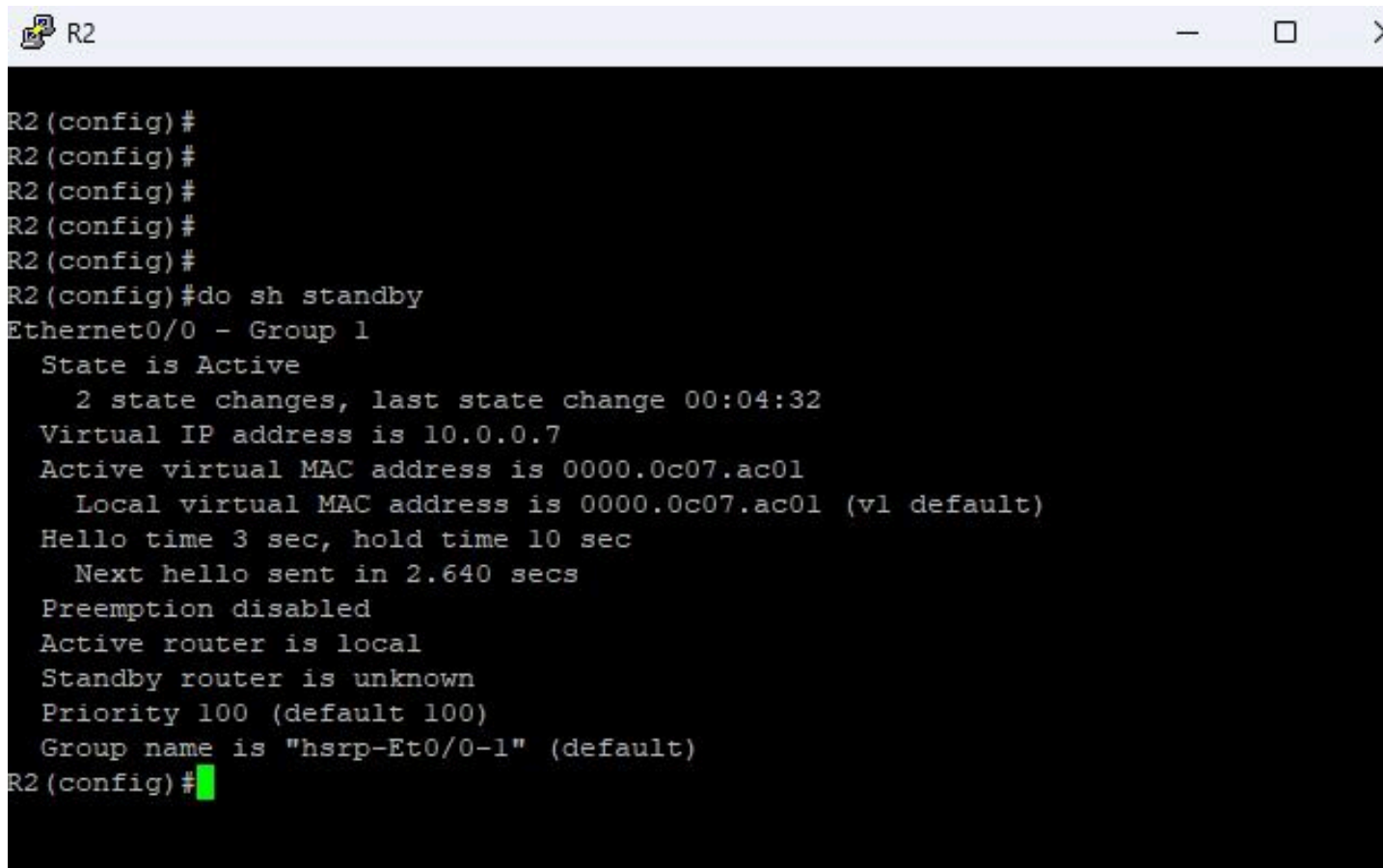
Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.7

C:\Documents and Settings\user>ping 20.0.0.2 -t

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Request timed out.
Request timed out.
Reply from 20.0.0.2: bytes=32 time=3ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=1ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
Reply from 20.0.0.2: bytes=32 time=2ms TTL=254
```



```
R2 (config)#
R2 (config)#
R2 (config)#
R2 (config)#
R2 (config)#
R2 (config)#do sh standby
Ethernet0/0 - Group 1
  State is Active
    2 state changes, last state change 00:04:32
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.640 secs
  Preemption disabled
  Active router is local
  Standby router is unknown
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R2 (config)#
```



```
R1
Group name is "hsrp-Et0/0-1" (default)
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
*Aug 15 15:28:02.487: %HSRP-5-STATECHANGE: Ethernet0/0 Grp 1 state Speak -> Standby
R1#sh standnby
      ^
% Invalid input detected at '^' marker.

R1#sh standby
Ethernet0/0 - Group 1
  State is Standby
    1 state change, last state change 00:00:14
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.976 secs
  Preemption disabled
  Active router is 10.0.0.2, priority 100 (expires in 10.448 sec)
  Standby router is local
  Priority 100 (default 100)
  Group name is "hsrp-Et0/0-1" (default)
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
```

```
R1
!
!
!
interface Ethernet0/0
 ip address 10.0.0.1 255.0.0.0
 ip nat inside
 ip virtual-reassembly in
 standby 1 ip 10.0.0.7
 standby 1 priority 110
 standby 1 preempt
 standby 1 track 1 decrement 30
 duplex auto
!
interface Ethernet0/1
 ip address 20.0.0.1 255.0.0.0
 ip nat outside
 ip virtual-reassembly in
 duplex auto
!
interface Ethernet0/2
 no ip address
 shutdown
 duplex auto
!
interface Ethernet0/3
 no ip address
 shutdown
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip nat inside source list 10 interface Ethernet0/1 overload
ip route 0.0.0.0 0.0.0.0 20.0.0.2
!
ipv6 ioam timestamp
!
!
access-list 10 permit 10.0.0.0 0.255.255.255
--More--
```

```
R2
!
interface Ethernet0/0
 ip address 10.0.0.2 255.0.0.0
 ip nat inside
 ip virtual-reassembly in
 standby 1 ip 10.0.0.7
 standby 1 preempt
 standby 1 track 1 decrement 30
 duplex auto
!
interface Ethernet0/1
 ip address 30.0.0.1 255.0.0.0
 ip nat outside
 ip virtual-reassembly in
 duplex auto
!
interface Ethernet0/2
 no ip address
 shutdown
 duplex auto
!
interface Ethernet0/3
 no ip address
 shutdown
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip nat inside source list 10 interface Ethernet0/1 overload
ip route 0.0.0.0 0.0.0.0 30.0.0.2
!
ipv6 ioam timestamp
!
!
access-list 10 permit 10.0.0.0 0.255.255.255
!
control-plane
!
!
```



```
R1
!
control-plane
!
!
!
!
!
!
!
!
!
!
line con 0
  logging synchronous
line aux 0
line vty 0 4
  login
  transport input none
!
!
end

R1#
R1#
R1#
R1#
R1#sh standby
Ethernet0/0 - Group 1
  State is Active
    2 state changes, last state change 00:06:00
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.744 secs
  Preemption enabled
  Active router is local
  Standby router is 10.0.0.2, priority 100 (expires in 8.048 sec)
  Priority 110 (configured 110)
    Track object 1 state Up decrement 30
  Group name is "hsrp-Et0/0-1" (default)
R1#
```

```
R2#
R2#
R2#
R2#
R2#
R2#
R2#
R2#
R2#sh standby
Ethernet0/0 - Group 1
  State is Standby
    4 state changes, last state change 00:05:18
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.904 secs
  Preemption enabled
  Active router is 10.0.0.1, priority 110 (expires in 10.912 sec)
  Standby router is local
  Priority 100 (default 100)
    Track object 1 state Up decrement 30
  Group name is "hsrp-Et0/0-1" (default)
R2#
```

```
R1
% Invalid input detected at '^' marker.

R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int e0/1
R1(config-if)#sh
R1(config-if)#end
R1#
*Aug 15 16:27:18.718: %TRACK-6-STATE: 1 interface Et0/1 line-protocol Up -> Down
R1#
*Aug 15 16:27:19.609: %SYS-5-CONFIG_I: Configured from console by console
*Aug 15 16:27:19.706: %HSRP-5-STATECHANGE: Ethernet0/0 Grp 1 state Active -> Speak
R1#
*Aug 15 16:27:20.720: %LINK-5-CHANGED: Interface Ethernet0/1, changed state to administratively down
*Aug 15 16:27:21.720: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to down
R1#
*Aug 15 16:27:31.283: %HSRP-5-STATECHANGE: Ethernet0/0 Grp 1 state Speak -> Standby
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
```

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#sh standby
Ethernet0/0 - Group 1
  State is Standby
    4 state changes, last state change 00:00:19
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.304 secs
  Preemption enabled
  Active router is 10.0.0.2, priority 100 (expires in 9.312 sec)
  Standby router is local
  Priority 80 (configured 110)
    Track object 1 state Down decrement 30
  Group name is "hsrp-Et0/0-1" (default)
R1#
R1#
R1#
R1#
R1#
R1#
```

```
R2#
R2#
R2#
R2#
R2#
R2#sh standby
Ethernet0/0 - Group 1
  State is Active
    5 state changes, last state change 00:00:49
  Virtual IP address is 10.0.0.7
  Active virtual MAC address is 0000.0c07.ac01
    Local virtual MAC address is 0000.0c07.ac01 (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.680 secs
  Preemption enabled
  Active router is local
  Standby router is 10.0.0.1, priority 80 (expires in 10.816 sec)
  Priority 100 (default 100)
    Track object 1 state Up decrement 30
  Group name is "hsrp-Et0/0-1" (default)
R2#
```